Article sampling, screening, summarising, and checking

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Material availability statement

We do not make available the full corpus of searched and obtained articles from which sampling occurred because it is copyrighted material, most of which is not used for stimuli within the experiment, and the case for providing it under academic fair use is therefore not clear. We also therefore do not make available the code for sampling, screening, summarising, and checking, because most of this code cannot be run without the corpus. Instead, we provide (1) summary statistics of the obtained corpus, (2) a full natural language description of the sampling, screening, summarising, and checking process, including all LLM prompts, and (3) the 110 articles in HTML format (85 of which are summarised) which would be necessary for replication of the experiment and are therefore justified to be shared under academic fair use. This includes the 100 articles used in the main experiment (termed specific articles in this document), and 10 additional articles that we used to test the effect of all articles that are due to disruptive environmental protest, not just articles that cover specific protests (termed general articles).

Technical implementation approach

All sampling, screening, summarising, and checking code was implemented in Python. All LLM prompts were conducted via Anthropic's Claude Python API, model version Claude Sonnet 4.0. Temperature was set to zero for maximum reliability; further, once a specific prompt had been executed once, the response was locally cached so the prompt was not re-executed when re-running code. All LLM prompts used in processes described in this document are included in full in the LLM prompts document also part of this Technical Materials repository (/doc/article_processing_llm_prompts.pdf).

Sampling and screening

Sampling and screening process overview

All articles resulting from the searching process (see main text Methods section, subsection "Selecting disruptive climate protest articles | Article search" were obtained as candidates for screening into the population of articles from which articles were randomly quota-sampled. Because screening uses resources (LLM tokens and human time), the screening and sampling procedure was not to screen all candidate articles and then randomly quota-sample from the population of screened-in articles. Rather, there was random sampling from all candidates, with each sampled candidate assessed against screening criteria and discarded if not screened in or required for quota. This continued until both of the following were complete: (1) 100 specific articles were screened in which met the specific-protest-related criterion, and which met the media source quotas for specific articles; (2) 10 general articles were screened in which failed that specific criterion but met the broader criterion for the analysis of all articles owing their existence to disruptive protest in general, and met the media source quotas for general articles. The quota for the 10 general articles was determined in the same way as for the 100 specific articles (see main text Methods section), but this required a general article from The Sun which did not exist in the candidate population. This quota slot was therefore taken by an article from The Mirror, as the highest readership proportion source which would otherwise would have been unrepresented among these 10 (readership proportion is 0.046). There was one minor and unavoidable deviation from the preregistered process for sampling and screening, detailed below.

All obtained articles are either non-relevant (picked up by search criteria but do not owe their existence to disruptive environmental protest), general (owe their existence to disruptive environmental protest), or specific (a subset of general, focusing on specific protest). The step-by-step algorithm for combined sampling and screening was as follows:

1. Randomly sample an article without replacement from the obtained corpus.
2. Assess article for format criteria (e.g., is article in English); if fail, return to Step 1.
3. Assess article for general criterion (i.e., in a counterfactual world identical to ours, except that environmental protesters never used disruptive methods, would this article still exist in a form where the primary message was hardly changed); if fail, return to Step 1.
4. Assess article for specific criterion, which is stricter (i.e., is the article's main focus on one disruptive environmental protest, or a small number of closely linked such protests); if pass, add article to specific sample unless specific quota for source is already met; if fail, add article to general sample unless general quota for source is already met.
5. Check all quotas; if all met, end process; if not, return to Step 1.

Because the full population of obtained articles was not sampled for screening, there is some uncertainty about the size of the full population of articles owing their existence to disruptive environmental activism (articles in either the specific or general categories). However, the population totals can be estimated as the *N* passing either general or specific criteria divided by the *N* sampled for screening multiplied by the *N* obtained through search. Table T1 contains the relevant summary statistics.

Table T1. Summary statistics for sampling process per source, with population estimates

Source	***N* obtained through search**	***N* sampled**	***N* sampled passing criterion for specific**	***N* sampled passing criterion for general (but not specific)**	**Estimated *N* articles annually owing existence to disruptive environmental activism**
BBC	177	80	32	9	45
Daily Mail	435	38	13	2	86
ITV	50	9	3	1	11
Metro	228	195	6	3	5
Mirror	61	15	5	2	14
Sky	106	31	10	1	19
Sun	19	17	6	0	3
Telegraph	103	16	6	2	26
The Guardian	675	107	13	17	95
The Times	66	25	7	2	12
TOTAL	1920	533	101	39	316

Note: All columns except the final one concern the sampling period (2025-10-07 to 2023-10-08).

LLM coding of format criteria coding

The format criteria are summarised in Section B.2 of the main Supplementary Material document. Articles were evaluated for these criteria by LLM. Because our experience from piloting was that the LLM classification almost never returned false positives for exclusion (e.g., classifying an English article as non-English or an ordinary news article as a letter to the editor), there was no manual checking of articles that were screened out for failing these criteria.

The later screening stages included manual checking, including retrospective checking of whether articles included by the LLM should in fact have been rejected due to formatting issues (see below). This prompted the addition of one non-preregistered format criterion: podcasts were excluded (via human checking rather than prompt modification). This affected only one searched and scraped “article”, from the Guardian.

LLM coding of general criterion coding

The first part of general coding was a binary decision by LLM as to whether the article met the criterion. The key part of the relevant prompt was solicitation of a decision: “In a

counterfactual world identical to ours, except that environmental protesters never used disruptive methods, would this article still exist in a form where the primary message was hardly changed?" A "yes" decision means an article passes our "general" criterion.

LLM coding of specific criterion coding

The first part of "specific" coding was a binary decision by LLM as to whether the article met the criterion. The key part of the relevant prompt (included in full in the prompt section below) was solicitation of a decision as to whether "The article's main focus is on one disruptive environmental protest, or a small number of such protests all closely linked in theme, time and protest method."

Human double-checking and manual override

Although the pilot indicated good reliability between LLM and manual screening (Section B.3 of the main Supplementary Material document), all article screening-in (for passing format checks, for being "general" and for being "specific") was manually double-checked and potentially overriden. It is reasonable to manually check screened in but not screened out articles, because incorrect screening in is far more severe than incorrect screening out for being non-relevant. That is because the random sampling approach anyway excludes most articles that could be screened in.

Because we double-checked and potentially overrode all LLM subdivision of "general" articles into "specific" or "non-specific", we did not rely at all on this LLM distinction. We included this LLM coding anyway and examine its reliability because (a) it was part of the pipeline we developed, which we might have chosen to have less oversight over, and (b) we were curious.

One reason for curiosity was because manual inspection of articles from the study period, prior to LLM-coding of them, suggested general changes in the way that disruptive environmental protest was covered in the UK media between the time period used for piloting (the two years before the study period) and the study period itself. Because of changes in the political context, this time-period transition roughly coincided with an increase in media focus on downstream consequences of relevant protest, such as government repression, severe legal consequences, and protest about repression of protest. This was anticipated to potentially make the boundaries between general and specific articles less clear to the LLM. In particular it was anticipated that there would be more articles that were not primarily about specific protests, but might briefly mention them, i.e. most appropriately coded as general but non-specific, but prone to be erroneously coded as specific.

After articles were LLM-coded as "general (non-specific)" or "specific", they were presented to a human coder, who coded them as "general (non-specific)", "specific", "neither", or "general but failing format criteria". The human coder was blind to whether the LLM had coded "general (non-specific)" or "specific", but knew that the LLM had coded one of those. The human-allocated code overrode the LLM-allocated code when they differed. Human blind coding was carried out only until the quotas for specific articles from each outlet were met, at

which point the automated process chain (see below) was only searching for general articles, so the human coder simply accepted or overrode LLM classification of "general (non-specific)".

The expectation of a greater proportion of non-specific articles in the final period compared to the pilot period was confirmed empirically: in the most recent pilot coding article batch, only 7% of general articles were non-specific, whereas for screened study-period articles, 28% of general articles were non-specific, odds ratio 5.2, 95% CI [2.2, 14.5], $p < 0.001$, Fisher's Exact Test. (There were some differences in coding procedures between these batches, but these were judged unlikely to be enough to create this difference.) Table T2 shows how the above-described human coding compares with the LLM coding.The reliability for the specific vs. general distinction was good, Cohen's $\kappa = .75$ (but note again that this LLM classification was not relied upon because it was always checked and potentially overriden).

Table T2. Human and LLM coding of general and specific

	Human coding				
LLM coding	**Specific**	**General (non-specific)**	**General but fails format criteria**	**Neither general nor specific**	**Sum**
Specific	93	5	2	5	105
General (non-specific)	7	25	1	7	40
Sum	100	30	3	12	145

Summarising and production of final HTML articles

Preregistered summary process

A summarisation process was triggered for all articles (general or specific) that had a word count greater than 350 words. The prompt for specific article summarising gave a target main text length of 300 words, instructed the text to focus only on the single most prominently reported protest (to enable coding of features of that protest for the analysis of effects of protest features), and to preserve the tone of the article in terms of the balance of positivity and negativity towards the protest. The prompt for general (non-specific) article summarising was less directional – only emphasising the maintenance of balance – because articles of this type were more varied and did not focus on a particular protest. Articles with fewer than 350 words were passed directly to the next steps of final article production, as described below. The article title and subheader (if present) were always kept as per the original. LLM summaries did not always strictly respect the word limit of 300 as per our word counts, resulting in a median article word count (across summarized and unsummarized articles) of 349.

Additional checks

The observation that articles from the sampling period were sometimes more complex than articles from the pilot period (see above section “Human double-checking and manual override”) might be expected to cause issues for the validated and preregistered summary process (see Supplementary Material section “LLM article summary validation”). That process included only limited checking and correction. Manual review indicated that this summary process did usually still produce good results. However, for some articles, the additional complexity seemed to make it more challenging for the summary process to maintain the original balance between expressed pro- and anti-protester sentiment, for example because such sentiment was included in a segment of the article focused on downstream consequences, which the summary process was more likely to remove. For this reason, further checking steps were added to the end of the preregistered summary process. These checks also included additional checks that the end product was a plausible UK mainstream media article. These checks sometimes resulted in further modifications to the article. These checks are fully described in the “Final article production” section below.

Image selection and inclusion and production of final HTML format

Image selection and inclusion was the same for summarised and non-summarised articles and for specific and general articles. Up to three images from the original article were used, distributed throughout the body of the text. In cases where an article contained more than three images, only the first three were used. The paragraph break in the text closest to the middle point (by word count) was identified. If there was only one image, it went there; if there were two, they went at the top of the body text and in the middle; and if there were three, they went at the top, middle, and bottom of the body text.

The final production articles, as directly inserted into the Qualtrics survey and as now publicly released, are in a basic HTML format. It begins with an outlet logo, followed by the title tagged <h1>, the subheader (if present) tagged <h2>, and then the body text, interspersed with images as described above, using the <figure> tag to collect images (hosted by the media source organisation) together with their captions and an image alt text.

The alt text was generated by the LLM’s image processing capabilities, with a prompt asking for a 50 word description of the image, describing exactly what was seen and discouraging guesses as to what it represented, but explaining the context of disruptive environmental protest. This alt text was generated for three reasons. Firstly, this enabled stimulus material to be released with a full description of images, which we do not release ourselves and which may not be available indefinitely. Secondly, this enabled the final article plausibility checks to be informed by LLM processing which was aware of the image content (as described below in the “Final article production” section). Thirdly, this enabled LLM to be used to determine whether pictures should count towards word-count-based article-feature coding (see Supplementary Materials section D). The image alt text would not have been displayed to participants unless they took unusual steps to examine it, but the image caption was displayed immediately below the images.

Final article production - all steps

All 110 screened and sampled articles (100 specific and 10 general) were subject to the following production process:

1. The preregistered summarisation process occurred as described above. Articles not requiring summarising because their main text was 350 words or less were passed directly to Step 5 below.
2. A summarisation check was solicited via LLM prompt, asking the LLM to compare the summary with the original and either pass the summary or identify problems of one or more of the following types:
    a. Consistency - whether key aspects such as overall tone and balance between description of different aspects of the protest were maintained.
    b. Form - the summary should not introduce problems with the form of the article making it implausible as a mainstream UK news article.
    c. New material - the summary should not introduce new material not present in the original.
3. A human check of every summarisation was conducted by reading first the original article, then the summarised version, then the output of the LLM summarisation check. The article could be passed (proceed to Step 5) or failed (proceed to Step 4); pass or fail was a human judgement based on combined assessment of original, summary, and LLM check.
4. The human used either or both of two correction procedures:
    a. Directly human-determined string replacement.
    b. LLM-conducted summary correction, in which the LLM was prompted with the original and summarised articles and an instruction for the specific correction necessary. The results of this correction were human checked and the correction instruction modified and re-run if the human judged necessary, iteratively if necessary.
5. A final plausibility check was conducted by LLM, focused only on issues that might make it implausible as a well-formed mainstream UK news article, for example non-sequiturs and spelling and grammar mistakes. Ironically, some such issues were present in original articles not summarised due to brevity, but were corrected so as not to distract participants. In contrast to all previous steps, this step also involved checking pictures and their captions, in the context of the article. The LLM judged pass or fail (with a reason); passing articles were automatically ready for production; failing articles were human assessed and if the human agreed with the reason for failure, the article returned to Step 4.

A goal of the summarising and checking process was to use as little human judgement as possible when determining specific text content. This was because of assumptions that, despite some human-detectable flaws, LLM might generally determine specific text more objectively and more in line with the original article. For that reason, directly human-determined string replacement was avoided as much as possible except for basic formatting issues. For only two articles was it used for other reasons; in both cases because of issues with article flow that the LLM struggled to correct (e.g., most natural placement of description of cause of death in an

obituary). For transparency, we provide the entire file of correction instructions (/articles/summarisation_corrections.tsv in these Technical Materials), and we summarise these corrections in Table T3. Note that balance corrections (where the initial summary was either more positive or negative in tone than the original) were required in each direction with roughly equal frequency. Of the 110 articles, 26 required correction of issues which were not relatively trivial formatting issues, i.e., balance or plausibility issues, with the latter defined as non-formatting issues which might interfere with the article's plausibility (see Table T3 for further definitions).

Table T3. Numbers of articles subject to corrections in Step 4

Correction reason	**Via additional LLM prompt**	**Via human-determined string replacement**
Balance issues		
Too anti-protester	8	0
Too pro-protester	9	0
Formatting issues		
Link material removal (e.g., "You can follow BBC Hampshire & Isle of Wight on Facebook, X (Twitter), or Instagram.")	0	18
Inappropriate paragraph breaking	0	1
Removal of photo credits from image captions (which were judged distracting)	0	51
Subheader related issues (e.g., repetition in body text)	0	4
Spelling or grammar	0	6
Plausibility issues		
Clarity (e.g., incongruity between title and body text because summarisation removed material referred to in title)	3	0
Flow (e.g., natural placement of statement of cause of death in obitury)	0	2
Photo-related (e.g., a selected photo seems out of place due to removal of material by summary)	3	0

Unattributed quote (summary made it unclear who was quoted, which is very rare in real articles)	1	0